

Week 7 Workshop Part 2 Solutions

Question 1

These results suggest some evidence against the weak-form EMH, because all of the variance ratios are below 1: $VR_2 = 0.9021$, $VR_3 = 0.8404$, $VR_4 = 0.7920$

Question 2

(a) A random walk model with drift model is $p_t = \mu + p_{t-1} + \varepsilon_t$ where $\mu \neq 0$.

(b) To derive (1), using $p_{t-1} = \mu + p_{t-2} + \varepsilon_{t-1}$ to give

$$\begin{aligned} p_t &= \mu + p_{t-1} + \varepsilon_t \\ &= \mu + (\mu + p_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \end{aligned}$$

This can be repeated using $p_{t-2} = \mu + p_{t-3} + \varepsilon_{t-2}$ to give

$$\begin{aligned} p_t &= 2\mu + \mu + p_{t-3} + \varepsilon_{t-2} + \varepsilon_{t-1} + \varepsilon_t \\ &= 3\mu + p_{t-3} + \varepsilon_{t-2} + \varepsilon_{t-1} + \varepsilon_t \end{aligned}$$

This can be repeated any number of times, say n times, to give

$$p_t = n\mu + p_{t-n} + \varepsilon_{t-(n-1)} + \cdots + \varepsilon_t$$

For simplicity, suppose that $n = t$

$$\begin{aligned} p_t &= t\mu + p_0 + \varepsilon_1 + \cdots + \varepsilon_t \\ &= t\mu + \varepsilon_1 + \cdots + \varepsilon_t \end{aligned}$$

as required, assuming that $p_0 = 0$. As shown in Week 7 seminar,

$$\begin{aligned} E(p_t) &= \mu \cdot t \\ V(p_t) &= \sigma^2 \cdot t \end{aligned}$$

and hence p_t is non-stationary.

(c) The correlogram of p_t shows that the acf is very close to one at the first few lags, it is decaying but very very slowly. This indicates that p_t is non-stationary.

